

**pundit lab**

## **Operating Instructions**

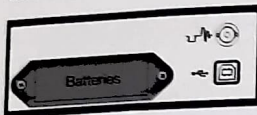
### **Pundit Lab / Pundit Lab<sup>+</sup> Ultrasonic Instrument**



**proceq**

## 2. Getting started

### Rear Panel



Connectors for:  
Oscilloscope

USB - PC / Power Supply

The Pundit Lab can be powered by batteries, by the mains or by the USB connection to a PC.

### Front Panel



The soft keys are context sensitive. The display icon shows the active function.

The lower right-hand key is for Power ON/OFF and also to "Cancel" a setting and return to the previous menu.

The navigation key is for scrolling through menus and for setting variable parameters.

### Transducer Connection

Connect the transducers to the front of the display unit using the BNC cables. If cables of different lengths are used, the longer cable should be connected to the transmitter.

**Note:** To avoid the possibility of electrical shock, the transducers must be connected before switching on and disconnected only after switching off.

### Basic Measurement



System Settings

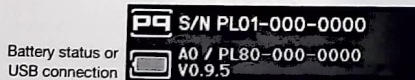
### Compound Measurement

Information

### Power ON

Press and hold the lower right-hand key for 3 seconds. The main menu and the type "Pundit Lab" or "Pundit Lab+" is displayed.

Clicking on the "Information" soft key brings up important device information.



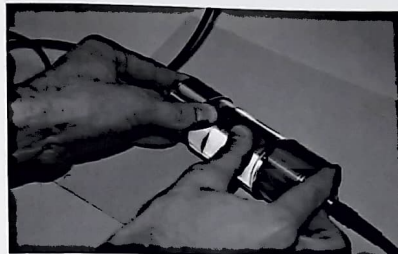
Serial number

Hardware version  
Firmware version

### Power Management - Power OFF

When operating on battery power the instrument goes into standby mode after 5 minutes of inactivity. After 30 minutes it shuts down completely. When powered by the USB connection, the display shuts down after 1 hour of inactivity. Press the lower right hand key (Power ON/OFF, Cancel) to recover from standby mode. To power off, press and hold the lower right-hand key for 3 seconds.

## Zeroing the Pundit Lab



The Pundit Lab should be zeroed using the calibration rod on a regular basis and in particular if the transducer frequency is changed or if the cables are changed. The expected calibration value ( $\mu$ s) is marked on the calibration rod.

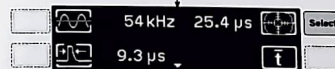
Couple the transducers to the calibration rod by applying couplant to the transducers and both ends of the rod and pressing firmly together as shown.



**Note:** When zeroing the exponential transducers the dedicated calibration rod (325 40 174) should be used.



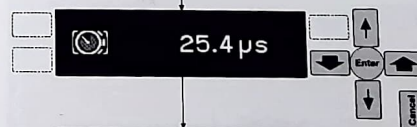
Select System Settings  
(See chapter 3 for more info)



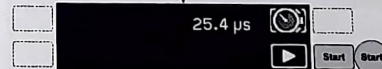
Check Tx/Rx matches transducer frequency. Pulse width requires no adjustment.  
Correction factor set to 1.0.  
Select Calibration.



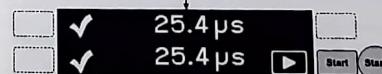
If necessary, select to enter the transit time as marked on the calibration rod.



**Variable parameter; set as follows:**  
Up/down keys - fine adjustment  
Left/right keys - coarse adjustment.  
Center key - enters the value and returns to the previous menu.  
Cancel key - cancels the input and returns to the previous menu.



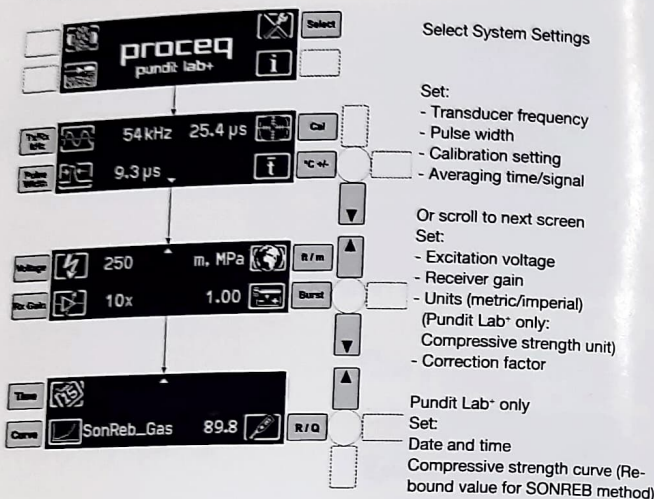
Press "Start" to begin the calibration sequence.



The final display shows the expected transit time and below it the measured transit time. This should match the value on the calibration rod.



### 3. System Settings



| Parameter                                | Presets (Click on soft key to scroll through) | Variable (Set with navigation key; see 2.1)                                     |
|------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------|
| Tx / Rx Frequency (kHz)                  | 24, 37, 54, 82, 150, 200, 220, 250, 500       | -                                                                               |
| Pulse width                              | Automatic                                     | 1-100 μs                                                                        |
| Calibration (See 2.1)                    | -                                             | 1-110 μs                                                                        |
| Averaging                                | Time                                          | Signal                                                                          |
| Excitation voltage (V)                   | 125, 250, 350, 500, AUTO                      | -                                                                               |
| Rx Gain                                  | Pundit Lab<br>Pundit Lab*                     | 1x, 10x, 100x, AUTO<br>1x, 2x, 5x, 10x, 20x, 50x, 100x, 200x, 500x, 1000x, AUTO |
| Units                                    | Pundit Lab<br>Pundit Lab*                     | ft / m<br>ft / m, MPa, N/mm2, psi, kg/cm2                                       |
| Correction factor                        | -                                             | 0.07 - 1.3                                                                      |
| Date and time (Pundit Lab*)              |                                               | Set time stamp.                                                                 |
| Compressive strength curve (Pundit Lab*) | Curves defined in PunditLink.                 | Enter a rebound value for SONREB curves.                                        |

#### Automatic Gain and Voltage Setting

The excitation voltage and the receiver gain can both be set to automatic. In this mode the Pundit Lab finds the optimum combination of the two parameters for a stable measurement.

#### Rx Gain 200x, 500x, 1000x

This feature is available in Pundit Lab\* only and replaces the need for an external amplifier (325 40 059) when using long cables or exponential transducers. When such high gain stages are selected it is highly recommended to use manual triggering with the waveform display.

#### Pulse Width

The pulse width is automatically set to the optimum value for the selected transducer frequency and typically requires no adjustment. For certain applications, it may however be adjusted if desired. (Refer to ASTM D 2845 - Standard Test Method for Laboratory Determination of Pulse Velocities and Ultrasonic Elastic Constants of Rock.)

It also allows non-standard transducers up to 500 kHz to be used with Pundit Lab: Pulse width "p" in microseconds (μs) to be entered here is calculated from the formula  $p = 1'000/(2f)$ , where  $f$  = transducer frequency in kHz



If a non standard value is set manually it is indicated by an asterisk.

#### Averaging Time/Signal



A number of measurements are made and the resulting transmission times are averaged. Time averaging provides the highest resolution of 0.1 μs.



Signal averaging provides greater stability for longer path measurements (e.g. several metres). In this case a number of measurements are made and the waveforms are averaged to determine the transmission time. The resolution in this mode is 0.5 μs.

#### Continuous/Burst Transmission



Continuous transmission continues transmitting until the "Stop" button is pressed.



Burst transmission sends out packets of pulses until a stable reading is obtained then stops automatically.

#### Correction Factor

Pulse velocity measurements are affected by several factors. These are well described in BS 1881 : Part 203 and it is recommended that the user refers to this document. Two key factors are the moisture content of the concrete and the temperature. The table below shows the correction factor that should be keyed in based on the recommendations made in the above standard.

| Temperature | Dry concrete        | Wet concrete        |
|-------------|---------------------|---------------------|
| 10°C - 30°C | 1.0 (No correction) | 1.0 (No correction) |
| 60°C        | 1.05                | 1.04                |
| 40°C        | 1.02                | 1.02                |
| 0°C         | 0.99                | 0.99                |
| -4°C        | 0.98                | 0.92                |

#### Units (Pundit Lab\* only)



m

kg/cm2

Selecting units opens a second screen where the user may select a compressive strength unit in addition to the imperial or metric units.

Date and Time (Pundit Lab\* only)



15:12:21

19.04.2011

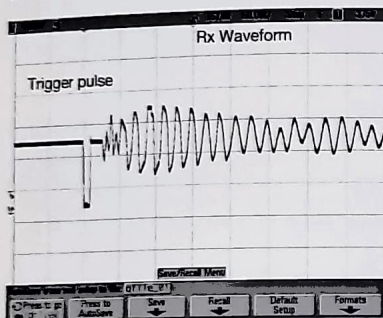
Used to provide measurements with a time stamp.  
Use the left and right keys to move between settings. Use the up and down keys to adjust the settings. Press enter to save, or cancel to return without saving.

#### 4. Viewing the Waveform

While carrying out the measurements described in the following chapter it is possible to view the received waveform. Pundit Lab offers three possibilities to look at the waveform.

##### Oscilloscope

Connect an oscilloscope to the connector on the rear panel. A typical waveform will look like this:

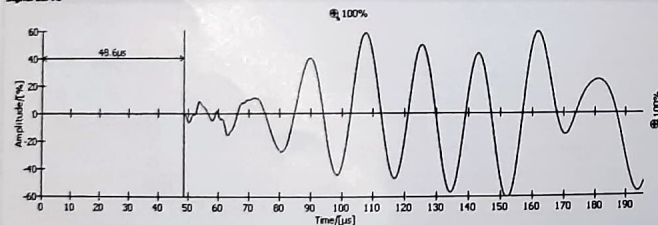


Both the trigger pulse and the received waveform are displayed.

##### Pundit Link - PC

The waveform may also be viewed on a PC or Laptop connected via the USB port. Please refer to the Pundit Link operating instructions for details (chapter 9).

##### Signal curve



##### On the instrument

The waveform may be directly viewed on the instrument. See next chapter for operating details.

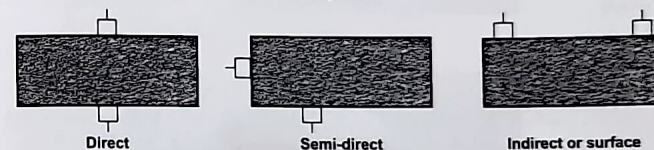
#### 5. Measuring with Pundit Lab

The Pundit Lab can be used for several applications including the following:

- Pulse velocity measurement
- Path length measurement
- Uniformity assessment
- Surface velocity measurement
- Crack depth measurement
- Estimating the dynamic elastic modulus of samples (with the shear wave transducers)
- Pundit Lab\* only. Estimating compressive strength using pulse velocity alone or in combination with a rebound hammer

##### Transducer arrangement

Three transducer arrangements are commonly used.



Wherever possible use the direct arrangement as this ensures the maximum signal transmission between the transducers. The semi-direct arrangement is less sensitive than the direct, but more sensitive than the indirect arrangement. The path length is the distance between the centers of each transducer.

The indirect method is particularly useful for determining crack depth, surface quality or in the case when only one surface is accessible.

##### 5.1. Preparation

Basic preparations are common to each application. The distance (path length) between the transducers should be measured as accurately as possible. It is very important to ensure adequate acoustic coupling of the transducers to the surface under test. A thin layer of couplant should be applied to the transducer and the test surface. In some cases it may be necessary to prepare the surface by smoothing it.

For compound measurements and uniformity testing a test grid should be drawn out on the surface.

Rebars affect the ultrasonic measurements as the signal will travel faster through the rebar than through the concrete. The location of rebars should be determined using a rebar locator such as Proceq's Profoscope and ultrasonic tests should be positioned so as to avoid them. BS 1881 Part 203 gives information on the effect of rebars on the expected results.

The standard measuring procedure is:

- Apply the couplant.
- Position the transducers.
- Perform the measurement.
- Reposition the transducers (compound measurements only).
- Save the result.



## 5.2. Basic Measurements

Basic measurements consist of a single measurement without the need to reposition the transducers. There are two basic measurements, depending on which parameter is known, either the path length or the pulse velocity.

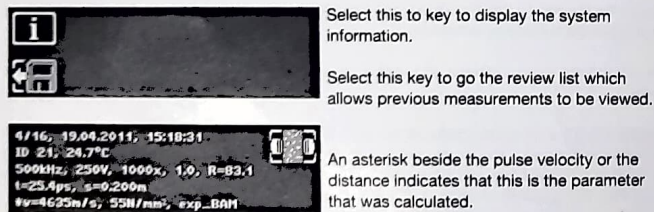


### Saving Measurements



### Review List (Pundit Lab\* only)

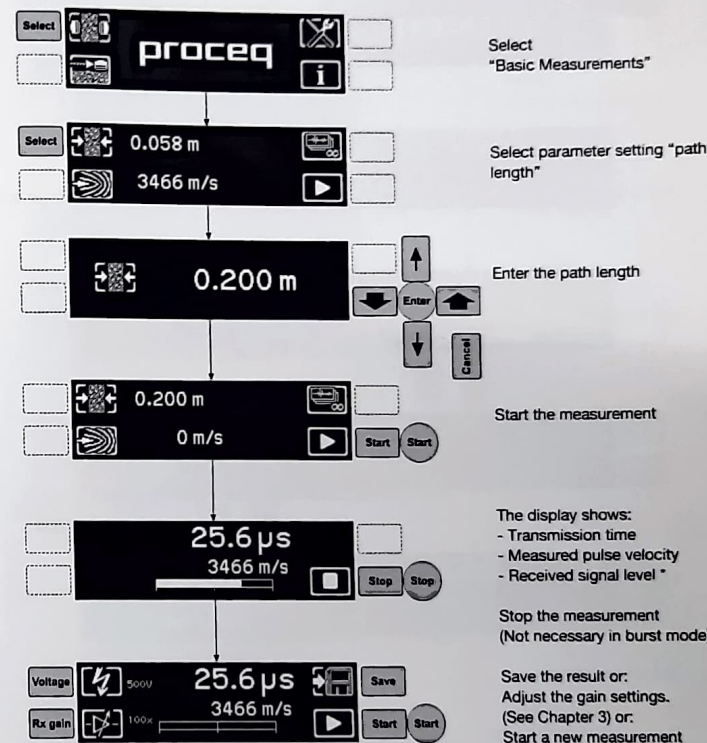
From the main menu, clicking on the system information key (See getting started) brings up a sub menu:



1st row: Measurement (e.g. 4 of 16) / Time stamp  
 2nd row: Measurement ID / Temperature  
 3rd row: Frequency / Voltage / Gain / Correction factor / Rebound value (SONREB only)  
 4th row: Transmission time / Distance  
 5th row: Pulse velocity / Compressive strength / Conversion curve

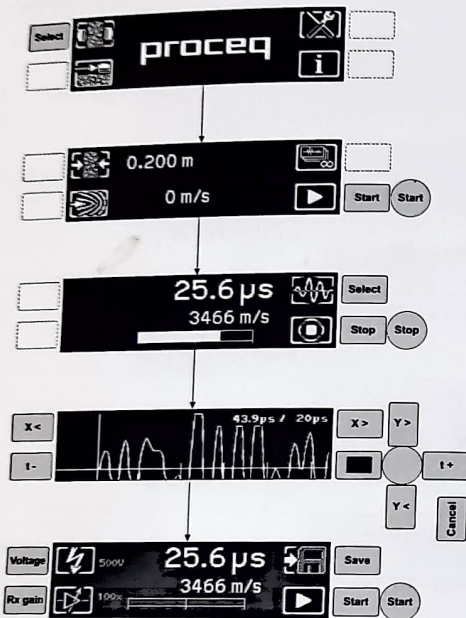
## 5.3. Basic Measurement - Pulse Velocity

In order to measure the pulse velocity it is necessary to measure the path length between the two transducers.



\* Best results are obtained when the received signal level is around 75%. Use the automatic gain and voltage settings (See Chapter 3) for optimized performance.

## 5.4. Basic Measurement – Waveform Display



### Waveform Controls

- Y <> Vertical zoom
- X <> Horizontal zoom
- t - + Manual trigger adjustment
- Stop/Start

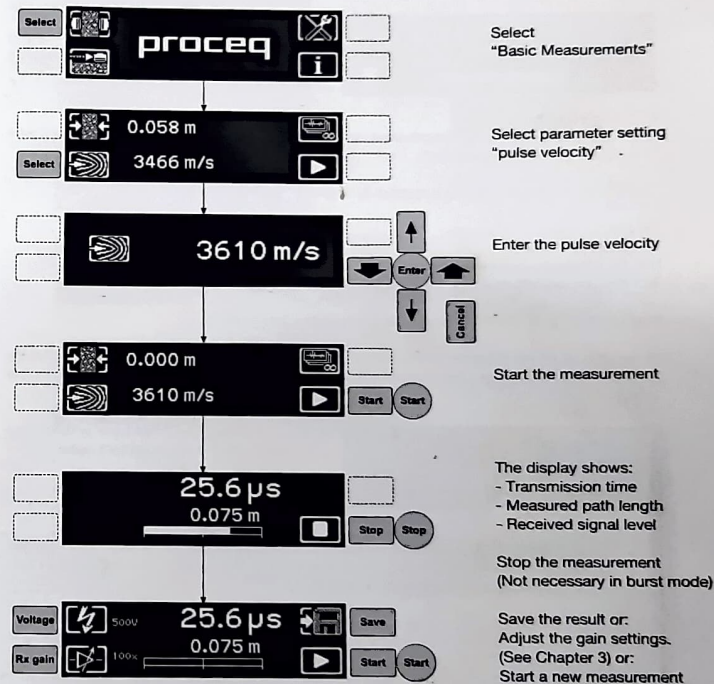
Note 1: Rx gain may need reducing for vertical zoom to have a noticeable effect.

Note 2: Once adjusted, trigger point does not reset automatically unless a new measurement is started in the menu "Start the measurement"

Note 3: The waveform is not saved, only the transmission time.

## 5.5. Basic Measurement – Path Length

If the pulse velocity of the material under test is known, it is possible to measure the path length between the transducers. The procedure is exactly the same as for measuring pulse velocity except that in this case the known pulse velocity is entered.



## 5.6. Basic Measurement – Compressive Strength (Pundit Lab<sup>®</sup> only)

Prior to carrying out this measurement, a conversion curve must be created in PunditLink and uploaded to the instrument. Up to five curves may be stored on the instrument. Select the appropriate curve and set the desired compressive strength unit. (See chapter 3 "System Settings".) Carry out a pulse velocity measurement as described above (5.3).



While the measurement is being carried out, or when the measurement has completed, clicking on the up or down keys switches between pulse velocity and compressive strength indication.

### SONREB Method

This method for measuring compressive strength combines the ultrasonic measurement with a rebound hammer measurement.

The format of the curve is: Compressive Strength  $f_c = aV^cS^c$

Where:

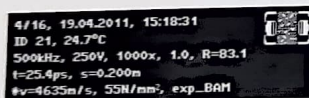
a, b and c are constants

V is the ultrasonic pulse velocity in m/s

S is the rebound value.

Many examples of SONREB curves can be found in the literature

In the settings menu (Chapter 3) the rebound value should be entered before carrying out the measurement. In this case the rebound value entered is saved along with the rest of the measurement data.



A rebound value of 83.1 has been entered here and saved with the measurement data.



**Note:** The rebound value may be a Q-value (SilverSchmidt) or an R-value (Original Schmidt). This is for the user to define, but it is always displayed in the review list as "R for rebound value".

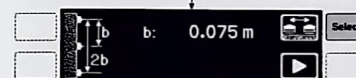
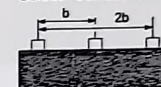
## 5.7. Compound Measurement – Surface Velocity



Select "Compound Measurements"



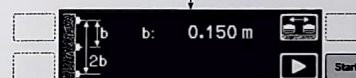
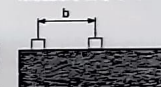
Select "Surface Velocity"



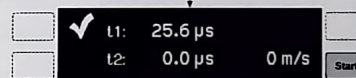
Select parameter setting "b".



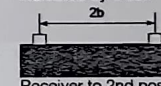
Measure and enter the distance "b"



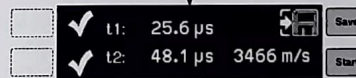
Transducers in first position – Start



"t1" is measured. Stable reading indicated by a tick



Receiver to 2nd position – Start



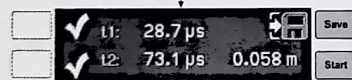
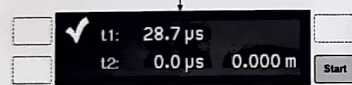
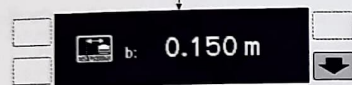
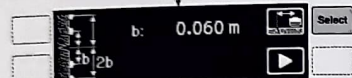
"t2" is measured. Stable reading is indicated by a tick and result screen is displayed.

The display shows:

- "t1"
  - "t2"
  - Surface velocity
- Save the result or press start to repeat the measurement.

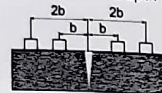


## 5.8. Compound Measurement – Perpendicular Crack Depth



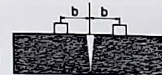
Select "Compound Measurements"

Select "Crack Depth"



Select parameter setting "b".

Measure and enter the distance "b"



Transducers in first position – Start

"t1" is measured. Stable reading indicated by a tick



Transducers in 2nd position – Start

"t2" is measured. Stable reading is indicated by a tick and result screen is displayed.

The display shows:

- "t1"

- "t2"

- Crack Depth

Save the result or press start to repeat the measurement.



**Note:** The crack depth measurement uses the method described in BS 1881:Part 203. For good results, the crack must be perpendicular to the surface and free of water or debris. The crack must be sufficiently wide to prevent the wave from propagating around it. Rebars close to the crack will also affect the result. If these conditions are not met, the crack depth will appear much smaller than it is.

## 5.9. Minimum Dimensions

Minimum dimensions are recommended for accurate test results.

### Minimum path length

100 mm for concrete with maximum aggregate size of 20 mm or less.

150 mm for concrete with maximum aggregate size between 20 mm and 40 mm.

### Minimum lateral dimensions of the sample

This is dependent on the transmission wavelength and the pulse velocity. The minimum path length should be equal to or greater than the transmission wavelength, or a severe reduction in the pulse velocity may be detected. The following table taken from BS 1881: Part 203 shows the minimum recommended lateral dimensions for the test object.

| Tx Frequency kHz                          | Pulse velocity<br>3500 m/s | Pulse velocity<br>4000 m/s | Pulse velocity<br>4500 m/s |
|-------------------------------------------|----------------------------|----------------------------|----------------------------|
| Minimum Lateral Dimensions of Test Object |                            |                            |                            |
| 24                                        | 146 mm                     | 167 mm                     | 188 mm                     |
| 54                                        | 65 mm                      | 74 mm                      | 83 mm                      |
| 150                                       | 23 mm                      | 27 mm                      | 30 mm                      |

## 5.10. Transducer Selection Guide

Generally, lower frequencies allow more depth of penetration. Higher frequencies allow better resolution in the measurements.

The inhomogeneities in concrete influence the propagation of an ultrasound pulse. This influence can be significantly reduced by choosing the frequency,  $f$ , such that the wave length,  $\lambda$ , is at least twice as large as the aggregate size.  $\lambda$  is given by:

$$\lambda = c/f,$$

where  $c$  is the pulse velocity (speed of sound) in concrete. The table below shows typical aggregate sizes and the corresponding maximum recommended frequency, respectively:

| c (m/s)             | 3500 |     |    | 4000 |     |    | 4500 |     |    |
|---------------------|------|-----|----|------|-----|----|------|-----|----|
| aggregate size (mm) | 8    | 16  | 32 | 8    | 16  | 32 | 8    | 16  | 32 |
| fmax (kHz)          | 219  | 109 | 55 | 250  | 125 | 63 | 281  | 141 | 70 |



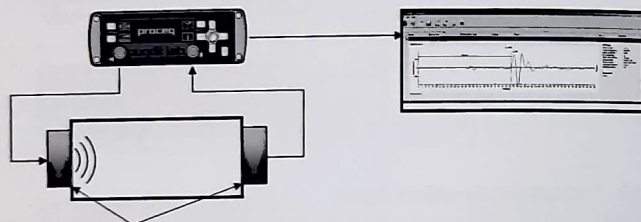
## 5.11. 250 kHz Shear Wave Transducers

When measurements with the 250 kHz shear wave transducers are performed, it is crucial to use the special shear wave coupling paste, otherwise shear waves cannot be properly transmitted into the object under test. The shear wave coupling paste is a non-toxic, water soluble organic substance of very high viscosity.

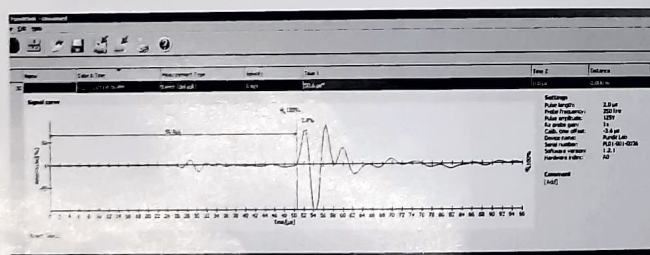
Furthermore, we highly recommend using Pundit Link's waveform display in order to manually locate the onset of the shear wave echo. Since the latter is always preceded by a relatively weak longitudinal echo (see Figure below), the transit time determined by Pundit Lab, would correspond to the longitudinal instead of the shear wave.

### Required steps before measurements can be performed

1. Put a small amount of shear wave coupling gel on the transducers.
2. Firmly press the transducers on either side of the 25  $\mu$ s calibration rod (Part No 710 10 028). Make sure that the coupling gel is properly distributed and that no air is trapped between the transducer and the calibration rod.
3. Connect the transducers to Pundit Lab.
4. Select the 250 kHz transducer from the list of supported transducers (see Pundit Lab manual chapter 3 for more details).
5. Zero the instrument as described in the Pundit Lab manual chapter 2.1.



Performing measurements with the 250 kHz shear wave transducers.



Typical echo signal obtained, with an experimental setup.

The first echo arrives at approximately 25.4  $\mu$ s and corresponds to the weak longitudinal wave. After 50.6  $\mu$ s the much stronger shear wave echo appears in the signal.

## 5.12. Transducer Holder Accessory



This accessory is particularly useful for compound measurements. (See chapter 5)

The individual transducer holders may be detached and used separately helping to reduce the physical strain when carrying out extensive measurements.

## 6. Technical Specifications

| Transit time Measurement          |                                                             |
|-----------------------------------|-------------------------------------------------------------|
| Range                             | 0.1 – 9999 $\mu$ s Auto ranging.                            |
| Resolution                        | 0.1 $\mu$ s                                                 |
| <b>Display</b>                    |                                                             |
| Display                           | 79 x 21 mm passive matrix OLED                              |
| <b>Transmitter</b>                |                                                             |
| Transmitter                       | Optimized energizing pulse 125 V, 250 V, 350 V, 500 V, AUTO |
| <b>Receiver</b>                   |                                                             |
| Selectable gain steps Pundit Lab  | 1x, 10x, 100x, AUTO                                         |
| Selectable gain steps Pundit Lab' | 1x, 2x, 5x, 10x, 20x, 50x, 100x, 200x, 500x, 1000x, AUTO    |
| Bandwidth                         | 20 kHz – 500 kHz                                            |
| Memory                            | Non volatile, > 500 measured values                         |
| Regional Settings                 | Metric and imperial units supported                         |
| <b>Power Supply</b>               |                                                             |
| Battery                           | 4 x AA batteries (> 20 hours continuous use)                |
| Mains:                            | 5v, <500mA via USB Charger                                  |
| PC                                | 5v, <500mA directly via USB cable                           |
| <b>Mechanical</b>                 |                                                             |
| Dimensions                        | 172 x 55 x 220 mm                                           |
| Weight                            | 1.3 kg (incl. batteries)                                    |
| <b>Environmental conditions</b>   |                                                             |
| Operating temperature             | -10° to 60°C (0° to 140°F)                                  |
| Humidity                          | <95% RH, non condensing                                     |
| IP Classification                 | IP42                                                        |